A methodological procedure and supporting tool for dedicated urban air mobility corridor design in the São Paulo metropolitan region

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Abstract—Integrating urban air mobility (UAM) into metropolitan airspace requires corridors respecting existing routes and airport flows. This paper presents a reproducible corridor-design procedure and dashboard for the São Paulo Metropolitan Region, applied to six- and twelve-vertiport networks on the Helicopter Special Routes (REH). Corridors 0.5 NM wide and 1,000 ft high, centered at 4,000 ft MSL between the REH and the Aircraft Special Routes (REA), are displaced 0.54–0.81 NM and lowered to 3,000 ft where they overlap. Volumetric overlap analysis shows the refinement lowered the corridor volume share inside other structures from 23.7% to 18.2%, mainly in the REA (8.1% to 3.1%). REH overlaps are thin (1.4%), but 14.0% lies in the Congonhas (SBSP) flow envelope derived from Flightradar24 records, largely from vertical allocation: the same paths in the REH have 2.8% of their centerline in that envelope and 0.3 SBSP movements a day crossing them, against 11.3% and 176 for the refined corridors. The 14.0% depends strongly on the envelope (2.0% for clustered streams), so overlap is read with observed traffic. The results are preliminary.

Keywords: urban air mobility, corridor design, airspace management, eVTOL operations, geospatial dashboard.

1. INTRODUCTION

Urban air mobility (UAM) must coexist with controlled airspace, airport flows and low-altitude traffic; structured corridors trade flexibility for predictability [1]. FAA and European concepts start from managed volumes or corridors [2, 3], which corridor studies translate into simulation geometry [4]. In the São Paulo Metropolitan Region (RMSP), such constructs must fit the Helicopter and Aircraft Special Routes (REH, REA) and the dense controlled airspace around Congonhas (SBSP) and Guarulhos (SBGR). Previous RMSP studies designed a vertiport and route network [5], estimated emissions avoided by eVTOLs [6] and applied reinforcement learning to strategic demand–capacity balancing [7].

This paper adds the corridor design: (i) a rule set justifying each dimension; (ii) a dashboard and workflow separating automated and analyst tasks; and (iii) reproducible verification by visual inspection, volumetric overlap with two airport-flow envelope representations and observed traffic on two networks. The scope is strategic design, not separation or capacity.

2. MATERIALS AND METHOD

2.1 Demand premises, vertiports and route graph

Terminals and demand premises follow [5]. Early demand is assumed to concentrate in high-income segments: helipads registered by ANAC in zones of the 2023 Metrô origin–destination

(OD) survey with per-capita income above R$ 5,000 were clustered by partitioning around medoids (k = 3), giving VP-001 to VP-003. The three heliports with the most recorded helicopter movements (BIMTRA) were added as VP-004 to VP-006 to form the initial stage, matching the expanded nine-node network of [5], and six heliports indicated by the São Paulo municipality for their regulatory status and FATO/TLOF dimensions form the twelve-vertiport stage [8]. OD matrices combine OD-survey trips (city taxi) with ANAC airport passengers (shuttle and transfer). The published matrix of [5] uses a 1% conversion factor and totals 349 daily operations for six vertiports; this paper uses the project scenario matrices [8], which apply 0.5% and total 198 and 574 daily operations, including SBSP, SBGR and Campo de Marte (SBMT). Demand sets the pairs, not corridor size.

REH and REA from the DECEA GeoAISWEB XML [9] were converted into volumes twice the published semi-width wide (REH full widths of 0.27 or 0.11 NM, corresponding to semi-widths of 250 or 100 m in uncontrolled or controlled airspace [10]; REA full widths of 1–3 NM), centered on the compulsory altitude or on the mid-point of the published band (the A→B band for REA published by direction), with a height equal to the band thickness; routes with only a compulsory altitude or a zero-thickness band take the mean thickness of their type (187 ft for 67 of 106 REH; 1,161 ft for 19 of 119 REA). An adapted Dijkstra search over the REH graph, with terminals linked to the nearest waypoints within 5 NM and barred as intermediate nodes, gives the shortest path of every terminal pair [5, 8, 11]: 9 terminals and 36 pairs initially, 15 and 105 in the intermediate stage. Altitudes are given in ft, horizontal distances in NM and volumes in km³, with configured metric values in parentheses.

2.2 Corridor dimensions and refinement rule

Figure 1 and Table 1 summarize the geometry. The REH are represented mostly at 3,000–3,500 ft MSL, and the design takes a nominal REA floor of 4,500 ft [10]. The 4,000-ft axis is the mid-point of this layer, about 1,500 ft above mean terrain, and the 1,000-ft height fills it, leaving 500 ft from axis to each band, half the conventional 1,000-ft minimum [12], and no buffer at the boundaries. In the GeoAISWEB data, however, 78% of the REA with published bands have lower floors (median 4,000 ft), so the layer is free only where no REA crosses the network. Each direction receives a 0.125-NM half-width lane (RNP 0.1 plus a 150-ft buffer [13]), giving 0.25 NM between tracks and 0.5 NM in total [4, 14].

A conflict is a pair of volumes that overlap both in plan and in altitude. In conflict regions, waypoints were displaced cross-track by d and lowered to 3,000 ft (2,500–3,500 ft), below the REA floor but within the REH band; lateral segregation from the REH being followed then requires $d > 0.25 + 0.135 = 0.385$ NM, the corridor half-width plus the larger REH semi-width (250 m), so the rule holds for every REH. The primary d = 0.54 NM (1.0 km) leaves a 0.15-NM margin (Figure 1c); 0.81 NM (1.5 km) was used where neighboring routes required more room, and a 1.08-NM (2.0 km) along-route component moved two turning points. Terminals stay fixed, and names, widths and heights are unchanged.

2.3 Empirical airport-flow envelopes

Flightradar24 records of December 2022 [15] (31 days, 30.4 million samples) were filtered to flights arriving at or departing from SBSP or SBGR, in final-approach, initial-climb and ground phases. Each airport has two parallel runways (SBGR 10L/28R and 10R/28L; SBSP 17R/35L and 17L/35R), grouped into one axis; the touchdown and lift-off samples (first and last zero

altitude) give the direction in use: 20,328 SBGR flights (80.2% on 10L/R) and 13,064 SBSP flights (65.1% on 17L/R). Altitudes are barometric, without QNH correction, and on-ground zeros take the field elevation (2,461 ft and 2,631 ft). For each side of the airport reference point (ARP), samples in that half-plane are assigned to stations every 0.1 NM of radial distance up to 10 NM, keeping each flight's nearest sample. The 5th (P5) and 95th (P95) altitude percentiles and the P95 absolute deviation from the extended runway axis are smoothed with cubic splines (smoothing 0.5). Each of the 404 cross-sections (4 sides × 101 stations) is centered on the runway axis, twice the lateral percentile wide and bounded vertically by P5 and P95, so its stored reference altitude is their mid-point. The lower limit rises from field elevation to 3,650–3,900 ft at 5 NM and 4,600–5,000 ft at 10 NM (Figure 2). A stream variant [16] applies k-means to resampled paths by runway and arrival/departure flow (up to six clusters per group, with at least 5% of flights and 1.5-NM mean-path separation). The 16 streams have 0.1-NM stations along their mean paths and P5–P95 lateral bands.

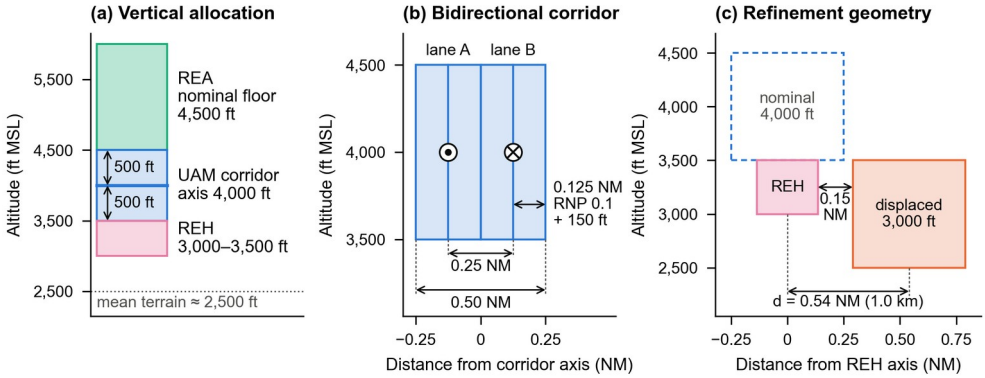


Figure 1. Corridor geometry: (a) vertical allocation; (b) two-lane cross-section, with traffic out of and into the page; (c) displacement of segments lowered to 3,000 ft beside an REH of the larger semi-width, 0.135 NM (250 m). Source: authors, based on [10], [13], [14].

Parameter	Value	Basis
Axis / height	4,000 ft MSL / 1,000 ft	Mid-point and full depth of the layer between the REH (3,000–3,500 ft) and the nominal REA floor (4,500 ft) [10]; 500 ft from axis to each band, half the 1,000-ft minimum [12]
Lane half-width	0.125 NM	RNP 0.1 (0.10 NM) plus a 150-ft buffer [13]
Track spacing / width	0.25 / 0.50 NM (generated: 0.27 / 0.52 NM)	Two opposite lanes [4, 14]; the earlier dashboard widened offsets by 11%, not widths; overlaps use this geometry
Refined altitude	3,000 ft MSL	Below the REA floor but inside the REH band, so segregation from the REH must be lateral
Cross-track offset d	0.54 NM (1.0 km); 0.81 NM (1.5 km) at 2–3 points	d > 0.25 + 0.135 = 0.385 NM (half-width + larger REH semi-width, 250 m [10]); 0.15-NM margin (0.14 NM as generated)
Along-route offset	1.08 NM (2.0 km) at 1–2 points	Moves turning points, once with a 0.27-NM (0.5 km) cross-track shift; resultant ≤ 1.35 NM

Table 1. Corridor parameters and their justification. Source: authors.

2.4 Dashboard operation and verification

The UAM Airspace Dashboard is a client-side HTML/JavaScript application (Leaflet 1.9.4 map, Three.js r128 3D view; Figure 3) [16]. It imports route CSVs, draws segments as boxes using the starting node's width and height, represents vertiports as cylinders, and generates parallel tracks in a local metric projection using offset segments and mitered interior waypoints while

preserving fixed endpoints; route CSVs are exported for downstream BlueSky processing [17]. It does not choose offsets or process Flightradar24 data.

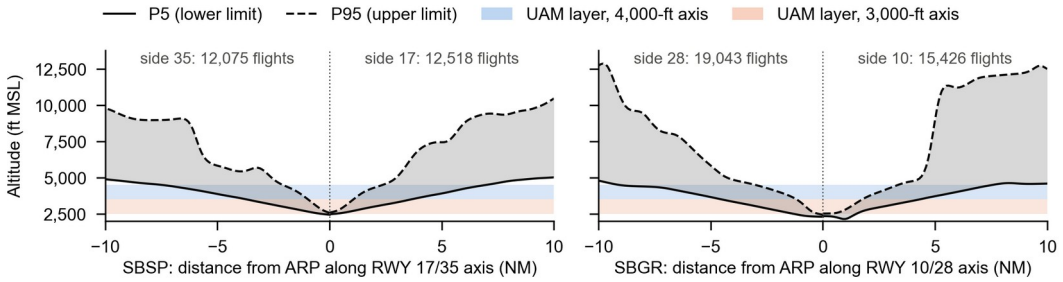


Figure 2. Vertical limits (smoothed P5 and P95) of the radial-method SBSP and SBGR airport-flow envelopes, plotted along the runway axes (positive toward 147° and 074°), December 2022, and the UAM layers. Source: authors; Flightradar24 [15].

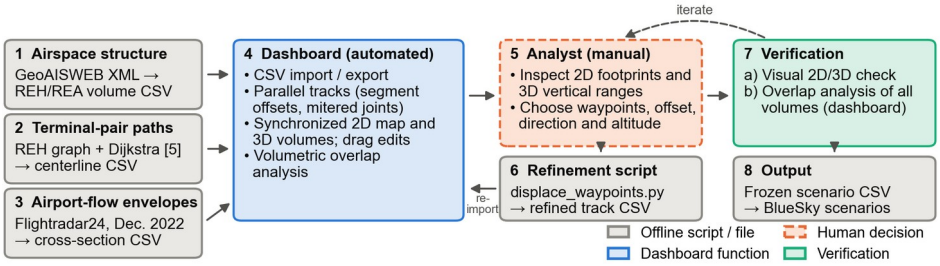


Figure 3. Workflow: offline scripts, dashboard functions, analyst decisions and verification. Source: authors.

Verification has two steps. In the visual step, the analyst loads the REH, REA, UAM and envelope files, scans the 2D volume footprints, compares the vertical extents of apparent overlaps in the transparent 3D view and records them as waypoint lists for the refinement script until no overlap is apparent. The automated step quantifies residual overlap: the dashboard's Overlap Analysis divides the plan into 0.027-NM (50-m) cells and, in each cell, merges the altitude ranges of each layer as drawn, so lanes and routes sharing a waypoint segment count once. It reports the corridor volume inside each layer, each segment's share inside other volumes, and splits overlap into parallel (pairs within 30°) and crossing parts; terminal climb and descent segments are reported apart. A test corridor height H (±H/2 around the tracks) can be set. As a reference, the original undisplaced paths, 99% within the REH they follow, were given the nearest REH band and width every 100 m (REH reference). Centerline shares are fractions of cruise path length inside the envelopes; traffic counts are distinct December movements a day whose tracks (densified to 50 m) cross a volume. The dashboard, tests, analysis and envelope scripts, XML converter, REH/REA files and both envelope variants [16] and the network code [11] are public; scenario CSVs are available on request; raw Flightradar24 data are licensed.

3. RESULTS

3.1 Refinement and verification

Visual inspection found two broad overlap regions between the 4,000-ft corridors and REA volumes, north-west and north-east (Figure 4a). The refinement changed 220 of 930 point

records (23.7%; 16 waypoints) in the six-vertiport case and 392 of 2,366 (16.6%; 18 waypoints) in the twelve-vertiport case, lengthening the track paths by only 0.11% and 0.37%.

The volumetric analysis (Table 2) shows the refinement lowering the share of corridor volume inside other volumes from 23.7% to 18.2% (24.9% to 19.6% in the twelve-vertiport case), mainly by cutting the REA share from 8.1% to 3.1%. Segments lowered to 3,000 ft lie in the REH band where they cross transverse REH, so the REH share rose from 0.6% to 1.4%, but these overlaps are thin: no segment has 10% of its volume inside an REH. The SBSP envelope dominates: it holds 14.0% of the corridor volume, and 19 segments 2.9–6.9 NM north-west of SBSP, under runway 17 arrivals and runway 35 departures, have at least 10% of their volume inside it, seven at least half. Overall, 23 of 59 segments have ≥ 10% of their volume in other volumes, 68% of the overlap being parallel, none found visually (Figure 4b).

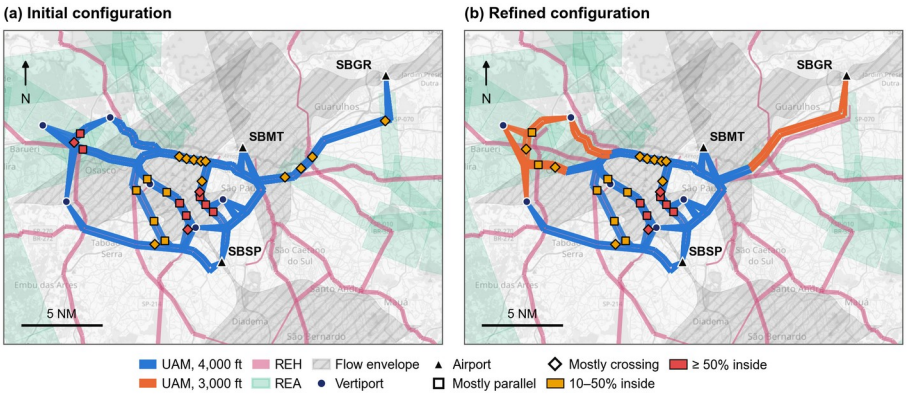


Figure 4. Six-vertiport network before (a) and after (b) refinement; markers: cruise segments ≥ 10% inside other volumes. Sources: GeoAISWEB [9], Flightradar24 [15]; basemap © OpenStreetMap contributors.

3.2 Corridor height and REH reference

Reducing the corridor height from 1,000 ft (±500 ft) to 600 ft (±300 ft) lowers the overlapping share only from 18.2% to 16.1%, because at the axis altitude 11.8% of the corridor footprint (14.1% for twelve vertiports) lies inside other volumes, mostly the SBSP envelope; an 800-ft corridor (±400 ft) at 4,000 ft would leave almost no REH overlap. The existing structures are almost clear of the observed flows: within 10 NM of the airports, 0.5% of the REH and 1.0% of the REA volume lie inside the envelopes, and in the REH reference 2.8% of the centerline lies in the SBSP envelope, against 11.3% for the refined corridors (Table 2). For six vertiports, the stream variant gives 2.0% (Table 2), almost all in departure streams, as the runway 17 final, 4–5 NM out, is a tube 0.02 NM wide. Track counts show the overlap is flown (Table 2): 130 SBSP movements a day follow one corridor segment on that final, and in one lane's box the same paths see 44 a day at 4,000 ft but 1.0 at REH altitudes. Vertical allocation, more than route geometry, is thus a major contributor to the SBSP overlap.

3.3 Network coverage and expansion

Expansion connects every terminal, raising the pair paths from 36 to 105, the unique centerline from 86.6 to 133.2 NM and the independent cycles (E − V + C) from 12 to 23. The mean pair-path length falls from 11.98 to 11.09 NM (−7.5%) mainly through composition: the 36 original

pairs change only to 11.90 NM, while the 69 new pairs, near the central vertiports, average 10.66 NM. Extra cycles add routing alternatives but do not show resilience.

Metric (cruise segments; initial → refined)	6 vertiports	12 vertiports
Share of corridor volume inside REH	0.6 → 1.4%	0.6 → 1.2%
inside REA	8.1 → 3.1%	7.6 → 2.4%
inside SBSP / SBGR envelope (radial)	14.7 / 0.4 → 14.0 / 0.0%	16.5 / 0.2 → 16.1 / 0.0%
inside SBSP envelope (stream)	2.1 → 2.0%	2.5 → 2.5%
inside any volume (radial; parallel part)	23.7 → 18.2% (68%)	24.9 → 19.6% (53%)
Segments ≥ 10% / ≥ 50% inside	26 / 10 → 23 / 7 of 59	32 / 14 → 29 / 9 of 79
Centerline in SBSP envelope (REH ref.)	11.7 → 11.3% (2.8%)	16.1 → 15.5% (1.9%)
SBSP movements a day inside (REH ref.)	176 → 176 (0.3)	262 → 262 (0.3)

Table 2. Volumetric overlap, centerline exposure and observed SBSP traffic for the two network stages (cruise segments; terminal climb and descent excluded). Source: authors; Flightradar24 [15].

4. DISCUSSION

Neither layer is free everywhere: most REA lie below 4,500 ft and the 3,000-ft layer lies in the REH band. Near SBSP the comparison exposes a trade-off: the dedicated layer segregates eVTOLs from helicopters but intrudes on the airport flows, which the REH largely avoid. That intrusion is highly sensitive to the traffic representation: the radial envelope screens terminal traffic broadly; the stream variant, resolving approach and departure paths by direction, sharply reduces the volumetric intersection; and flight tracks, an independent check of historical exposure, show that narrow, densely used approach paths still cross the corridors. Volumetric overlap and traffic exposure should therefore be assessed jointly. These findings motivate evaluating a hybrid that keeps the 4,000-ft axis with an 800-ft corridor away from the airports, reroutes around the REA and, within about 7 NM of SBSP, descends to REH altitudes or shares the REH under ATC coordination. Further limitations are the REH/REA representation, the reduced vertical spacing, mean terrain, RNP widths without containment analysis and one month of tracks that are not protected surfaces. Future work will iterate the geometry and simulate the networks in BlueSky for demand–capacity balancing [7] and deconfliction.

5. CONCLUSIONS

This paper documents a reproducible corridor-design procedure and dashboard for the RMSP. The refinement reduced REA overlap but left thin REH crossings, while 14.0% of the corridor volume remains inside the radial SBSP envelope, a share sensitive to the envelope representation; observed tracks confirm that approach paths cross the corridors. The REH reference has much lower centerline exposure and traffic, motivating evaluation of a hybrid near SBSP. The assessment is preliminary and does not establish an operationally certified design.

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